Whole Grain Roll-2 oz

Bread/Grain

Iowa Gold Star Recipe
Healthier Kansas Recipe 107

Ingredients	100 Servings		____Servings		Directions
	Weight	Measure	Weight	Measure	
White whole wheat flour	3 lb 12 oz	3 qt 2 ¼ cups			1. Place flour, dry milk, sugar, yeast and salt in mixer bowl. 2. Blend with dough hook for approximately 2 minutes on low speed. 3. Add oil and blend for approximately 2 minutes on low speed. 4. Add first amount of water to the dry ingredients. If dough appears too stiff, add up to 1 cup extra water per 100 servings. More or less water can be added as needed. Mix for 1 minute on low speed or until all water is mixed with the dry ingredients. 5. Knead dough on medium speed for 8-10 minutes until dough is properly developed. Use the gluten stretch test: At 8 minutes, stretch the dough like a rope. If it breaks, continue to mix 2 additional minutes. 6. Form dough into 2 lb balls and spray with non-stick cooking spray or cover with plastic wrap. Once all dough balls have been formed, begin shaping. 7. Form rolls from dough by pinching off 2 oz pieces and shaping. Place rolls in rows of 7 across and 10 down on sheet pans (18" x 26" x 2") which have been lightly coated with non-stick cooking spray. 8. Place in a warm area (about 90°F) until double in size, approximately 45-60 minutes. 9. Bake until lightly browned: Conventional oven: 400°F for 18-20 minutes Convection oven: 350°F for 12-14 minutes Turn pans half-way through baking time to promote even baking. Recommended internal temperature for baked whole wheat rolls is 196-198°F.
All-purpose flour, enriched*	3 lb 8 oz	3 qt ¾ cup			
Dry milk, instant non-fat	7 oz	1 ¾ cup			
Sugar, granulated	11.5 oz	1 ¼ c + 3 T			
Yeast, instant, dry	3 oz	½ cup			
Salt	1.75 oz	2 Tbsp 2 tsp			
Vegetable oil*	13 oz	1 2/3 cup (+ up to 1 cup if dough is stiff)			
Water (calculate water temperature by subtracting the temperature of the dry ingredients from 145°F)		2 qt ½ cup			
Non-stick cooking spray		As needed			

					10. Rolls will have a better appearance if lightly sprayed with non-stick cooking spray when they come out of the oven. Notes: • Dough made with whole wheat flour will absorb more water and requires increased rising/proofing time. Mixing time is less because the bran from the whole grain cuts through developing gluten strands with increased mixing, resulting in a product with low volume. Even an additional one minute mixing time with whole grains can make a difference in quality. Figuring water temperature so that the final dough temperature is about 80°F is essential for quality bread products.

*USDA Foods

Serving Size	1 Serving Provides	Yield
1 roll	2 oz eq grains.	12.5 lb dough

Nutrients Per Serving

Calories	170	Sodium	200 mg	Vitamin A	2 IU
Total Fat	4 g	Carbohydrates	29 g	Vitamin C	0.11 mg
Saturated Fat	<1 g	Dietary Fiber	2.5 g	Calcium	34 mg
Trans Fat	0 g	Protein	5 g	Iron	1.4 mg
Cholesterol	<1 mg	Calories from Total Fat	23%	Calories from Saturated Fat	3 %